MAT 1013: Discrete Mathematics

Session 5: Introduction to Proofs

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Activity sheet: Introduction to Mathematical Proofs

For each problem, use the indicated proof technique. Clearly state any assumptions and justify each step.

- Integers: $\mathbb{Z} = \{\cdots, -2, -1, 0, 1, 2, \cdots\}$
- Natural Numbers: $\mathbb{N} = \{1, 2, 3, \cdots\}$
- Non-negative integers: $\{0, 1, 2, \cdots\}$

A. Direct Proofs (Existence Statements)

1. Prove that there exists an integer n such that $n^2 + n + 41$ is odd.

2. Prove that there exist an integer n such that $2^n > n^2$.

3. Prove that there exist a real number x such that $\sqrt{x^2} \neq x$

B. Direct Proofs (Universal Statements)

1. Definition: For a non-negative real number a (i.e., for $a \geq 0$), $\sqrt{a}$ is defined as the unique non-negative real number such that its square is a.
Prove that for any real number x, $\sqrt{x^2} = |x|$, where $|x|$ is the absolute value of x.

2. Prove that for every real number x, $x^2 + 1 \geq 2x$.

C. Counterexamples

Determine whether each statement is true or false. If false, provide a counterexample.

1. For every integer n $n^2 + n + 41$ is prime.

2. For all real numbers x and y, $\sqrt{x + y} > \sqrt{x} + \sqrt{y}$.

D. Proof by Contrapositive

1. Prove that if n^2 is even, then n is even.

2. Let $a, b \in \mathbb{Z}$. Prove that if ab is odd, then both a and b are odd.

3. Prove that if $n^2 + 1$ is odd, then n is even.

E. Proof by Contradiction

1. Prove that there is no largest integer.

2. Prove that $\sqrt{3}$ is irrational.

3. Prove that there are infinitely many prime numbers.

4. Prove that $log_2(3)$ is irrational.

F. Proof by Cases

1. Prove that for every integer n, $n^2 + n$ is even.

2. Prove that for all $x \in \mathbb{R}$

$$|x| + |x - 1| \geq 1.$$

G. Mathematical Induction

1. Prove that for every $n \geq 1$,

$$1 + 3 + 5 + \cdots + (2n - 1) = n^2.$$

2. Prove that for every $n \geq 1$,

$$1 \cdot 2 + 2 \cdot 3 + \cdots + n(n + 1) = \frac{n(n + 1)(n + 2)}{3}.$$

Challenge Problems

Attempt these after completing the main worksheet.

1. Prove that for every integer n, $n^3 - n$ is divisible by 6.
2. Prove that if n is an odd integer, then $n^2 - 1$ is divisible by 8.
3. Prove that for every integer $n \geq 1$, $2^n > n$.
4. Prove that among any three consecutive integers, one is divisible by 3.
5. Prove that there is no smallest positive rational number.

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